Fiscal Year:	FY 2021	**Task Last Updated:**	FY 06/29/2021
PI Name:	Turek, Fred W Ph.D.		
Project Title:	Effects of Spaceflight on Gastrointestinal Microbiota in Mice: Mechanisms and Impact on Multi-System Physiology		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	60201-2919	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	06/01/2015	**End Date:**	05/31/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 5/31/2022 per NSSC information (Ed., 9/23/21) NOTE: End date changed to 5/31/2021 per F. Hernandez/ARC and NSSC information (Ed., 7/17/20) NOTE: End date changed to 5/31/2020 per F. Hernandez/ARC and NSSC information (Ed., 6/14/19) NOTE: End date changed to 5/31/2019 per NSSC information (Ed., 6/7/18)		
Key Personnel Changes/Previous PI:	June 2021 report: Co-Investigator Stefan Green has moved to Rush University from University of Illinois, Chicago (UIC).		
COI Name (Institution):	Green, Stefan Ph.D. (Rush University) Keshavarzian, Ali M.D. (Rush University Medical Center) Forsyth, Christopher Ph.D. (Rush University Medical Center) Vitaterna, Martha Ph.D. (Northwestern University) Jiang, Peng Ph.D. (Northwestern University)		
Grant/Contract No.:	NNX15AL05G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	This proposal in response to NNH14ZTT002N NASA Research Announcement (NRA) “Research Opportunities for Flight Experiments in Space Biology” is intended to examine the potential role(s) of disruption of microbial communities in the gut (dysbiosis) in mammalian adjustment to the space environment. Many challenges to mammalian physiological homeostasis present in the space environment are known to create dysbiosis. Similarly, immune or inflammatory changes, as well as metabolic alterations, observed with time in spaceflight may be attributable in part to dysbiosis. A series of experiments in mice will be carried out to define the mechanisms by which time in space, diet, and host genotype interact to impact the composition of the gut microbiota, and how dysbiosis relates to gene expression (by RNA-seq) and physiology in serum, colon, ileum, spleen, liver, and fat as well as the sleep/wake cycle and feeding behavior. Extensive use of ground based studies will support the analysis and interpretation of spaceflight data. Five projects at three institutions will focus on different aspects. Project 1 (University of Illinois at Chicago, Principal Investigator (PI: SJ Green) will carry out metagenomic sequencing of microbiota from fecal samples and cecal contents (including from the tissue sharing opportunity); Projects 2 and 3 (Rush University Medical Center, PIs: A Keshavarzian and C Forsyth) will use gene expression, physiological and histologic measures of gastrointestinal and immune function, respectively; while Projects 4 and 5 (Northwestern University, PIs: FW Turek and MH Vitaterna) will focus on metabolism, sleep, and circadian behavioral organization. Dr. Turek (Northwestern) will provide Principal Investigator (PI) oversight. These studies are in alignment with NASA Research Announcement Research Emphases 1)“Spaceflight Omics Studies” and 3)“Understanding How Complex Organisms Adapt to Spaceflight.”
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal will examine the impact of the space environment on the population structure of the intestinal microbiota of mice, and how multiple physiological systems involving the gastrointestinal (GI), immune, metabolic, circadian, and sleep systems, known to be affected by the microbiota, are impacted by the space environment. While our overall working hypothesis is that the space environment and associated stressors will have a negative impact on the microbiota (i.e., dysbiosis), and that this in turn will negatively impact multiple physiological systems known to be adversely affected by spaceflight, our integrated analysis of gene expression and multiple tissue/organ specific biomarkers of tissue/organ injury will also allow us to uncover the adverse effects of spaceflight due or not due to dysbiosis. However, the impact of the proposed studies extends beyond understanding the mechanisms at play in the unique stresses of spaceflight. The proposed studies will elucidate mechanisms underlying interactions between GI, immune, metabolic, and sleep functions and specifically the role of the microbiota in these interactions. This knowledge has tremendous potential for guiding development of dysbiosis-targeted interventions for disorders in all of these systems.
Task Progress:	The ultimate goal of this study is to determine the effects of spaceflight over time on an interconnected set of physiological systems: 1) the gut microbiota; 2) gastrointestinal function; 3) immune function; 4) metabolism; and 5) sleep and circadian behavioral organization. Rodent Research 7 (RR7) compares mice of two different genotypes (inbred strains): C57BL/6J (the most widely-used mouse strain) and C3H/HeJ. The study also included comparisons of two different durations of spaceflight (~25 and ~75 days). In addition to the Flight group, three ground control groups were also part of the study: Basal (representing the pre-launch state), Vivarium (standard vivarium housing for the same duration of time as flight), and Ground (same habitat in the International Space Station Environment Simulator, ISSES). Twenty mice (10 of each strain) were included in each experimental group. All mice completed the study without significant health issues. Samples were obtained from a total of 80 mice in 2018. This year, an additional 40 Ground control mice were exposed to the identical ISSES conditions and schedule, as a test of the impact of the habitat light spectrum. This provided a side-by-side comparison of the light spectra of the original Flight ("white") and Ground ("blue-enriched") lighting. All mice completed the study without significant health issues. In addition, using shotgun metagenomic sequencing, we characterized the microbiome in fecal and environmental samples from the 2018 study animals. The sequencing data were processed using the NASA GeneLab metagenomics processing pipeline (https://), and the taxonomic abundance and microbial gene content data were derived from sequencing reads via both read-alignment-based and assembly-based approaches. Preliminary findings based on only the read-based taxonomic abundance data suggest mouse(host)-strain-dependent changes in the fecal microbiome during spaceflight. Using principal component analysis (PCA) followed by Permutational Multivariate ANOVA (PERMANOVA) on isometric-log-ratio-transformed species abundance data, we found significant differences in the fecal microbiome composition between B6 and C3H mice throughout the mission ($P < 0.0001$). By mission day 25, clear effects of the habitat (i.e., the ISS hardware and environment used by Flight and Ground groups vs. the standard caging used by Vivarium and Basal groups; $P < 0.0001$), spaceflight ($P < 0.0001$), as well as habitat × mouse strain ($P = 0.015$) and spaceflight × mouse strain ($P < 0.0001$) were observed. This pattern was persistent till the end of the study (i.e., day 75). These findings are consistent with our hypothesis that the host genetic background influences the effects of the space environment on the gut microbiome.
Bibliography Type:	Description: (Last Updated: 07/16/2025)
Articles in Peer-reviewed Journals	Afshinnekoo E, Scott RT, MacKay MJ, Pariset E, Cekanaviciute E, Barker R, Gilroy S, Hassane D, Smith SM, Zwart SR, Nelman-Gonzalez M, Crucian BE, Ponomarev SA, Orlov OI, Shiba D, Muratani M, Yamamoto M, Richards SE, Vaishampayan PA, Meydan C, Foox J, Myrrhe J, Istasse E, Singh N, Venkateswaran K, Keune JA, Ray HE, Basner M, Miller J, Vitaterna MH, Taylor DM, Wallace D, Rubins K, Bailey SM, Grabham P, Costes SV, Mason CE, Beheshti A. "Fundamental biological features of spaceflight: Advancing the field to enable deep-space exploration." Cell. 2020 Nov 25;183(5):1162-84. Review. https://doi.org/10.1016/j.cell.2020.10.050 ; PMID: 33242416 , Nov-2020